# DPP — DEPARTMENTAL POLICY AND PROCEDURE

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| TITLE: | CARE OF NEWBORN WITH RESPIRATORY DISTRESS SYNDROME |
| DEPARTMENT: | NEONATAL INTENSIVE CARE UNIT |
| APPLIES TO: | NICU PHYSICIANS AND NURSES |

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| PREPARED BY: | MS. KRISTINE P. MAGNETICO | UNIT QUALITY REPRESENTATIVE | |
| | MS. SHATHA MOHAMMED | NICU HEAD NURSE | |
| REVIEWED BY: | MEDICAL EXECUTIVE COMMITTEE | | |
| | DR. ABDULKARIM IBRAHIM | HEAD OF NICU | |
| | MS. MERRY JOY GALLEBO | NURSING QUALITY COORDINATOR | |
| | MR. KHALID GHURMULLAH ALGHAMDI | QPS DIRECTOR | |
| APPROVED BY: | MS. HAIFA SAAD MISFER | DIRECTOR OF NURSING | |
| | DR. RAKAN KHALIL ALGHAMDI | AED FOR MEDICAL SERVICES | |

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CONTENTS


| | | |
|---|---|---|
| 1. PURPOSE | 4. POLICY | 7. REFERENCE/CROSS-REFERENCES |
| 2. DEFINITION & ABBREVIATION | 5. PROCEDURE | 8. ATTACHMENT |
| 3. RESPONSIBILITY | 6. MEASUREMENT | 9. REVISION CLAUSE |

## 1. PURPOSE

1.1. To provide guidelines to physician on how to manage sick newborn with respiratory distress syndrome.

1.2. To emphasize the nursing care to be rendered to patients with RDS.

## 2. DEFINITION & ABBREVIATION

2.1. **Respiratory Distress Syndrome (RDS)** – is a condition of insufficiency of pulmonary surfactant in which there is an increasing respiratory distress commencing at, or shortly after birth and increasing in severity which manifest tachypnea, cyanosis and grunting and is mainly confirmed to preterm patients.

2.2. **Tachypnea** – rapid breathing between 60-120 breaths per minute.

2.3. **Retinopathy of prematurity (ROP)** – is a disease of the eye affecting premature newborn patients generally having received intensive neonatal care, which oxygen therapy is used on them due to the premature development of their lungs.

2.4. **Grunting** – a deep pitched sound which is heard from newborn suffering from RDS.

2.5. **Cyanosis** – blue discoloration of the skin during breathing efforts.

2.6. **PDA** – Patent Ductus Arteriosus

2.7. **CLD** –Chronic Lung Disease

2.8. **NICU** – Neonatal Intensive Care Unit

## 3. RESPONSIBILITY

3.1. **Physicians (Consultant, Specialists and Residents)**

3.1.1. Assess and evaluate patients with respiratory distress syndrome.

3.1.2. Provide a written order of the procedures to be done in the file

3.2. **NICU Nurse (Head Nurse, Charge Nurse and Staff Nurse)**

3.2.1. Responsible to in monitoring the patient's respiratory rates and other vital signs.

3.2.2. Document all the procedures to be done to manage/treat RDS.

## 4. POLICY

4.1. All newborn patients who develop respiratory distress must be carefully placed under observation in Nursery and will be subjected for admission in Neonatal Intensive Care Unit as indicated according to the newborn's clinical condition and physician's written order.

4.2. Patient with RDS may be required to be intubated to ensure full lung expansion and for the administration of prophylactic surfactant.

4.3. Patient with RDS must be closely observed with:

4.3.1. Clinical Presentation

4.3.1.1. Tachypnea (fast breathing)

4.3.1.2. Tachycardia (fast heart rate)

4.3.1.3. Chest wall retractions (recession)

4.3.1.4. Expiratory grunting

4.3.1.5. Nasal flaring

4.3.1.6. Cyanosis

4.3.1.7. Rising carbon dioxide concentrations in the blood

4.3.1.8. Prolonged cessation of breathing (apnea)

4.4. Diagnosis must be made by the clinical picture and the chest X-ray, which demonstrate as follows:

4.4.1. Decreased lung volumes (bell-shaped chest)

4.4.2. Absence of thymus (after about 6 hours)

4.4.3. A small (0.5 – 1mm), discrete, uniform infiltrate appearance that involves all lobes of the lung

4.4.4. Infiltrate will outline the larger airways passages which remain air-filled.

4.4.5. In severe cases, infiltrates become exaggerated until the cardiac borders become in apparent (a white-out appearance).

4.5. Surfactant can be given immediately the baby is born (Prophylactic Treatment) or several hours later when the diagnosis of RDS has been established (Rescue Treatment). Refer to DPP-NICU-0008 Surfactant Administration.

4.6. CPAP is given to improve oxygenation by increasing the functional residual capacity of the collapsed alveoli. CPAP is indicated in patients with RDS who have a Pa02 persistently below 7kPa (50-60 mmHg) despite an increase in their inspired oxygen to 50% or above. However, in babies above 2kg birth weight, or 32 weeks' gestation, a period of CPAP can be tried as a response may avoid the need for ventilation.

4.7. Patients in CPAP must be closely observed and it must only be continued if patients show adequate respiratory effort, and are maintaining satisfactory arterial/venous blood gases.

4.8. If the arterial oxygenation remains unacceptably low after 2 doses of surfactant, then the infant has failed to respond to surfactant. Such patients are relatively rare and a cause should be carefully assessed for associated sepsis, pulmonary hypertension, pneumothorax or pulmonary interstitial emphysema which requires mechanical ventilation.

4.9. Blood gases monitoring should be done. Firstly, as guide for the ventilator support and secondly to minimize the risk for ROP.

4.10. **Prevention of Infection:**

4.10.1. Initial blood investigation of patients with RDS should include a septic screen (CRP and Blood Culture).

4.10.2. Antibiotic therapy should be initiated while waiting for laboratory confirmation that infection was absent.

4.11. **Fluid Balance**

4.11.1. RDS delays the onset of the normal postnatal diuresis and edema often appears after 24-48 hours. Careful attention to fluid replacement therapy needs to be given to avoid fluid overload, which contributes to the complications of PDA and CLD.

4.12. **Blood Transfusion to replace Blood Loss:**

4.12.1. Preterm infants with RDS easily become anemic from blood sampling and replacement blood transfusion is often necessary for anemia and/or hypotension.

4.13. Nursing care must be given to patients with RDS which include:

4.13.1. **Intensive Care Monitoring**

4.13.1.1. Successful management of RDS requires careful monitoring and attention to the details of intensive care support which preterm babies require. This includes frequent monitoring not only of blood gas concentrations but blood glucose, heart rate, respiration temperature and blood pressure.

4.13.2. **Suctioning**

4.13.2.1. Suctioning is essential to maintain airway patency and a necessary component of care for CPAP patients and is frequently carried out as a routine nursing practice. It removes airway secretions, thereby preventing obstruction, atelectasis and decreased lung compliance while optimizing oxygenation and ventilation.

4.13.2.2. Complications associated with frequent suctioning however should be avoided such as bacteremia, hypoxemia, tachycardia and bradycardia, atelectasis, systemic hypertension and raised intracranial pressure.

4.13.3. **Physiotherapy**

4.13.3.1. Chest physiotherapy must be ordered by physicians when it comes to timing, frequency and duration since routine chest physiotherapy is not recommended in neonatal RDS because it contravenes the minimal handling concept.

4.13.4. **Positioning of patients with RDS**

4.13.4.1. The supine position allows a good access to the patient for intubation,

cannulation and other procedures. It also affords the opportunity for clear visual assessment of chest wall movement and symmetry and the site of the umbilical venous catheter.

4.13.4.2. Keep the head in constant slight extension which can help stabilize and improve air entry.

4.13.4.3. Supine positioning may not optimal in terms of oxygenation and energy expenditure for which prone position is better. Head flattening and excessive hip flexion are also a problem.

4.13.4.4. It is encouraged to have a balance between flexion and extension in a variety of positions prone, supine and lateral.

## 5. PROCEDURE

5.1. The physician and the nurse will assess the patient with respiratory distress.

5.2. The physician will provide a written order of the procedures to be done to manage or treat RDS.

5.3. The nurse will verify and carry out the doctor's written order.

5.4. Identify the patient correctly.

5.5. Perform hand hygiene and wear appropriate PPE according to the procedure to be done.

5.6. The nurse will proceed in performing the procedure to manage RDS according to physician's written order and policy.

5.6.1. Observation and monitoring of vital signs

5.6.2. Oxygen therapy

5.6.3. Antibiotic therapy

5.6.4. Surfactant administration

5.7. The nurse provides the appropriate nursing care for the newborn patient with respiratory distress syndrome according to physician's order and management.

5.8. Document all the procedures done in the patient's medical file.

## 6. MEASUREMENT

6.1. 100% of the NICU staff is knowledgeable and adhere to this policy and procedures.

## 7. REFERENCES /CROSS- REFERENCES

7.1. PMSH-DPP-NICU-0014(v3): Care of Newborn with Respiratory Distress Syndrome, 28/05/2023, p. 1-4

7.2. CBAHI Standards 3rd edition, 01/01/2016, NICU.10, NICU.12 p.126-127

7.3. MOH KSA Guidelines for Neonatal Care

7.4. www.thoracic.org/patient/patient-resources/respiratory-distress

## 8. ATTACHMENT

8.1. None

## 9. REVISION CLAUSE

9.1. Changes to the hospital policy and procedure template based from the Al-Baha Health Cluster logo, font face and design.